



Complete Technical Report

All changes, infrastructure, and features built — Dec 25, 2025 to present

Report Date:	March 13, 2026
Period:	December 25, 2025 — March 13, 2026
Total Database Tables:	71
Total Decision Receipts:	63,543
Total Evaluation Cycles:	747,422
Classification:	Internal — Not for distribution

Table of Contents

1. Executive Summary
2. Timeline — Dec 2025 to Jan 2026
3. Timeline — Feb 2026 (First Half)
4. Timeline — Feb 2026 (Second Half)
5. Timeline — Mar 2026 (First Week)
6. Timeline — Mar 2026 (Second Week)
7. Database — B2B Client Tables
8. Database — Sandbox Tables
9. Database — Alert System Tables
10. Database — Governance Receipts (PQC-signed)
11. Database — 5 Compliance Modules
12. Database — Decision Engine & Vetos
13. Database — Backtesting Phase 0
14. Database — Trading (Real & Paper)
15. Database — Users & Configuration
16. API Endpoints — B2B External
17. API Endpoints — Sandbox
18. API Endpoints — Alerts
19. API Endpoints — Compliance Modules

- 20. Web Pages (omnixquantum.net)
- 21. Documents Generated
- 22. Summary in Numbers

1. Executive Summary

This report documents every significant change, feature, and infrastructure component built into OMNIX from December 25, 2025 through March 13, 2026. It covers database tables (71 total), API endpoints, web pages, compliance modules, and generated documents. All database row counts are queried live from the local PostgreSQL instance at generation time.

Key numbers: 747,422 evaluation cycles processed, 63,543 PQC-signed receipts generated, 25,386 vetos logged, 2,237 real Kraken trades recorded, 5 B2B clients registered, 4 vertical demos live.

2. Timeline — Dec 2025 to Jan 2026

Foundation of the public website and bot stabilization

- Created the public website (omnixquantum.net) with separate commercial and institutional pages
- Added contact sections with WhatsApp, FAQ, competitor comparison table
- Connected frontend with Railway backend correctly
- Improved bot AI response speed (timeout optimization and model selection)
- Upgraded primary AI model to Gemini 2.5 Flash
- Configured dynamic Track Record day counter (no more hardcoded number)
- Implemented dashboard caching for faster loading
- Fixed navigation and unknown route redirects
- Added leadership team section to institutional page
- Configured Railway deployment with Python + Node.js build pipeline

3. Timeline — Feb 2026 (First Half)

Repositioning as Decision Governance + Multi-vertical demos

- OMNIX officially repositioned as 'Decision Governance Infrastructure' — not a trading bot
- Added voice interface for hands-free governance interaction
- Implemented AI response compression (contextual: institutional vs casual tone)
- Built Insurance Governance Demo — underwriting with governance checkpoints
- Built Credit/Lending Governance Demo — credit risk evaluation
- Built Energy Trading Governance Demo — natural gas, crude oil, solar, wind, LNG, electricity
- Built Biotech/Clinical Trial Governance Demo — real ClinicalTrials.gov NCT data
- Updated bot tone to be more institutional, less servile
- Improved AI greeting detection and natural language handling
- Created mentor packages with personalized draft messages and resources

4. Timeline — Feb 2026 (Second Half)

B2B infrastructure, security, and investor documents

- Built B2B External Governance API with API key authentication (X-API-Key header)
- Implemented RBAC (Role-Based Access Control) — regular clients vs admin with separate permissions
- Added payload encryption at rest using Fernet (AES-128-CBC + HMAC-SHA256)
- Built 5 Compliance Modules aligned with NIST AI RMF, ISO/IEC 42001, and EU AI Act
- Module 1: Risk Mapping — use case classification, impact scoring, stakeholder mapping
- Module 2: Measurement & Monitoring — checkpoint metrics, drift detection
- Module 3: Human Oversight — override logging with PQC-signed audit trail
- Module 4: Incident Management — incident reporting, review, and corrective actions

- Module 5: Governance Reporting — period-based reports with content hashing
- Generated research paper PDF with reproducibility section
- Created 3-tier security documentation: public, institutional, and internal
- Removed Ivan Guzman from all files — Harold Nunes as sole visible Founder
- Launched public deployment on Railway + custom domain omnixquantum.net
- Generated multiple investor PDF documents and mentor packages
- Clarified PQC communication rules (no FIPS 204, use 'NIST-standardized')
- Implemented API key rotation and database backup documentation

5. Timeline — Mar 2026 (First Week)

Eureka Dubai preparation and presentation materials

- Created complete pitch deck with professional slides for Eureka Dubai (March 18)
- Generated Business Model Canvas PDF
- Created executive report for investors
- Prepared presentation script with real operational data
- Built Q&A; simulation guide (tough investor questions with prepared answers)
- Added Brazil financial regulations context for channel partner Richard (Blue Mesh)
- Updated AI Risk Analyst report with independent validation
- Fixed Black Swan detection classification logic
- Made all website and contact links clickable in generated reports

6. Timeline — Mar 2026 (Second Week)

8-checkpoint architecture + real-time data pipeline

- Expanded governance pipeline from 6 to 8 checkpoints: added CP-0 (Signal Integrity) and CP-7 (Temporal Coherence)
- Built interactive Sandbox for B2B clients — test evaluations without API key
- Built configurable Alert System — Slack, email, webhook notifications per client
- Added per-client threshold customization — each company adjusts their own checkpoint thresholds
- Fixed all pages to consistently show 8 checkpoints (Energy, Biotech, Credit, Insurance demos)
- Connected Railway public API to return all live metric fields (decisions_blocked, capital_preserved_pct, system_uptime_days, verticals_demo)
- Made system_uptime_days calculate automatically each day from Jan 15, 2026
- Eliminated all remaining hardcoded values in the web (22,000+, 98.5%, -1.5%)
- All web pages now refresh metrics every 60 seconds from Railway production API

7. Database — B2B Client Tables

Table	Rows	Columns (from DB schema)
b2b_clients	5	id, client_id, api_key_hash, name, email, role, is_active, created_at, last_seen_at
client_thresholds	0	id, client_id, checkpoint_id, threshold, updated_by, updated_at, created_at

b2b_clients stores registered API clients with hashed keys and RBAC roles. client_thresholds allows per-client checkpoint threshold customization with audit trail.

8. Database — Sandbox Tables

Table	Rows	Columns (from DB schema)
sandbox_sessions	3	id, session_id, client_id, name, description, evaluation_count, created_at, expires_at, last_eval
sandbox_evaluations	6	id, evaluation_id, session_id, client_id, asset, domain, signals, decision, decision_score, check

The sandbox allows potential clients to test the governance engine without authentication. Sessions expire automatically. Each evaluation generates a full 8-checkpoint trace.

9. Database — Alert System Tables

Table	Rows	Columns (from DB schema)
alert_configs	2	id, alert_config_id, client_id, alert_type, channel, enabled, config, created_at, updated_at
alert_events	1	id, event_id, alert_config_id, client_id, alert_type, channel, trigger_data, delivery_status, deliver

10. Database — Governance Receipts (PQC-signed)

Table	Rows	Columns (from DB schema)
decision_receipts	63,543	id, receipt_id, timestamp_utc, asset, decision, veto_chain, policy_version, engine_version, prev
exit_governance_receipts	78	id, receipt_id, position_id, symbol, exit_reason, regime, should_exit, gate1_threshold_verdict, g

Every governance decision generates a cryptographically signed receipt using Dilithium-3 (NIST-standardized post-quantum algorithm). Receipts form a SHA-256 hash chain for tamper detection. Exit governance receipts track the 3-gate exit pipeline with regime-adjusted thresholds.

11. Database — 5 Compliance Modules (NIST AI RMF / EU AI Act)

Table	Rows	Module — Columns (from DB schema)
governance_risk_map	0	Module 1 — Risk Mapping: id, client_id, use_case, classification, impact_financial, impact_operational, risk_score, last_updated
governance_metrics	0	Module 2 — Measurement: id, client_id, checkpoint_id, approval_rate, block_rate, avg_score, v
governance_drift_log	0	Module 2 — Drift Detection: id, client_id, signal_name, baseline_stats, current_stats, drift_score
governance_overrides	0	Module 3 — Human Oversight: id, client_id, receipt_id, decision_before, decision_after, justification
governance_incidents	0	Module 4 — Incidents: id, client_id, incident_id, severity, title, description, related_receipt_id, st
governance_incident_reviews	0	Module 4 — Reviews: id, incident_id, reviewer, findings, corrective_actions, reviewed_at, creat
governance_reports	0	Module 5 — Reporting: id, client_id, period_start, period_end, report_type, content_json, conte

Five additive governance modules aligned with NIST AI RMF, ISO/IEC 42001, and the EU AI Act. Each module introduces dedicated PostgreSQL tables and REST endpoints. All override operations generate PQC-signed audit trails.

12. Database — Decision Engine & Vetos

Table	Rows	Columns (from DB schema)
shadow_trade_events	747,422	id, created_at, symbol, market, intended_action, intended_quantity, intended_entry_price, inter
trading_veto_log	25,386	id, created_at, veto_type, engine_stage, symbol, market, user_id, trade_id, blocked_capital, cu
filter_calibration_metrics	0	id, calculated_at, period_start, period_end, veto_type, symbol, market, total_vetoes, vetoes_wi
shadow_trade_outcomes	50	id, shadow_event_id, calculated_at, price_at_veto, price_after_1h, price_after_4h, price_after

13. Database — Backtesting Phase 0

Table	Rows	Columns (from DB schema)
backtest_phase0_results	225	id, eval_key, asset, eval_hour, decision, veto_chain, checkpoints_passed, actual_pnl_usd, cou
backtest_phase0_signals	258	id, eval_key, asset, eval_hour, probability_score, risk_exposure, signal_coherence, trend_pers
backtest_phase0_ohlcv	0	id, asset, candle_time, open, high, low, close, volume, fetched_at

14. Database — Trading (Real & Paper)

Table	Rows	Columns (from DB schema)
kraken_real_trades	2,237	id, txid, refid, trade_time, type, subtype, aclass, asset, amount, fee, balance, amount_usd, phase
paper_trading_trades	119	id, user_id, symbol, side, quantity, entry_price, exit_price, profit_loss, profit_pct, strategy, status
paper_trading_balances	1	id, user_id, balance_usd, btc_balance, eth_balance, total_trades, winning_trades, losing_trades
paper_trading_daily_reports	1	id, user_id, report_date, balance_usd, daily_pnl, period_pnl, roi_pct, win_rate_pct, sharpe_ratio

Table	Rows	Columns (from DB schema)
trades	0	id, user_id, symbol, side, quantity, price, total, fee, order_id, exchange, status, executed_at

15. Database — Users & Configuration

Table	Rows	Columns (from DB schema)
users	14	user_id, telegram_id, username, first_name, last_name, language_code, is_premium, is_admin
user_settings	4	user_id, username, risk_profile, trading_limits, protection_settings, notification_preferences, all
conversations	107	id, user_id, user_message, ai_response, language, created_at
system_config	0	id, config_key, config_value, description, updated_at

16. API Endpoints — B2B External (authenticated with X-API-Key)

Method	Endpoint	Function
POST	/api/governance/evaluate	Run 8-checkpoint evaluation, return PQC-signed receipt
GET	/api/governance/receipts	List client receipts (isolated by client_id)
GET	/api/governance/schema	Documentation of accepted signals
POST	/api/governance/admin/clients	Create client and generate API key (admin)
DELETE	/api/governance/admin/clients/<id>	Deactivate client (admin)
POST	/api/governance/admin/clients/<id>/rotate	Rotate API key (admin)
GET	/api/governance/admin/clients/<id>/thresholds	Get client thresholds (admin)
PUT	/api/governance/admin/clients/<id>/thresholds	Set per-client thresholds (admin)
DELETE	/api/governance/admin/clients/<id>/thresholds	Revert to defaults (admin)
GET	/api/governance/metrics	Public real-time governance metrics
GET	/api/verify/<receipt_id>	Public receipt verification

17. API Endpoints — Sandbox (no authentication)

Method	Endpoint	Function
POST	/api/governance/sandbox/sessions	Create test session
GET	/api/governance/sandbox/sessions	List sessions
GET	/api/governance/sandbox/sessions/<id>	Get session details
DELETE	/api/governance/sandbox/sessions/<id>	Delete session
POST	/api/governance/sandbox/evaluate	Run evaluation without API key
GET	/api/governance/sandbox/schema	Signal schema documentation

18. API Endpoints — Alerts

Method	Endpoint	Function
PUT	/api/governance/alerts/config	Create alert configuration
GET	/api/governance/alerts/config	List client alert configs
POST	/api/governance/alerts/test	Test an alert
DELETE	/api/governance/alerts/config/<id>	Delete alert config

19. API Endpoints — Compliance Modules

Module	Endpoints	Function
Risk Mapping	POST/GET /api/governance/risk-map	Create and list risk maps per client
Monitoring	GET /api/governance/metrics/live	Live approval/block statistics
Drift Detection	POST /api/governance/drift/detect	Detect signal drift vs baseline
Human Oversight	POST /api/governance/overrides	Log decision overrides with PQC signature
Incidents	POST/GET /api/governance/incidents	Report and track governance incidents
Incident Reviews	POST /api/governance/incidents/<id>/review	Submit incident review findings
Reporting	POST/GET /api/governance/reports	Generate and retrieve governance reports

20. Web Pages (omnixquantum.net)

Page	Status	Description
Commercial Landing	Live	Public-facing page with live metrics refreshing every 60 seconds
Institutional Page	Live	Track Record, PQC overview, competitor comparison, verifiable metrics
Credit/Lending Demo	Live	8-checkpoint governance applied to credit risk evaluation
Insurance Demo	Live	8-checkpoint underwriting governance with risk assessment
Energy Trading Demo	Live	8-checkpoint energy trading governance (gas, oil, solar, wind, LNG)
Biotech/Clinical Demo	Live	8-checkpoint clinical trial governance (real ClinicalTrials.gov data)

All pages use the useLiveMetrics hook which refreshes data every 60 seconds from the Railway production API. Fallback values are used only when Railway is unreachable. System uptime days are calculated dynamically from Track Record start date (January 15, 2026).

21. Documents Generated

- Pitch deck for Eureka Dubai GCC 2026 (March 18 presentation)
- Business Model Canvas PDF
- Executive Report for investors
- Research Paper with reproducibility section
- Personalized mentor packages (multiple recipients)
- Q&A; Simulation — tough investor questions with prepared answers
- Presentation script with real operational data
- AI Risk Analyst Report (independent validation)
- 3-tier Security Documentation (public / institutional / internal)
- Internal Security Audit v1.0
- Validation questionnaires for industry professionals (multiple recipients)
- LinkedIn thought-leadership posts (English & Spanish)

22. Summary in Numbers

Metric	Value	Notes
Total database tables	71	PostgreSQL on Railway
Evaluation cycles	747,422	shadow_trade_events rows
PQC-signed receipts	63,543	Dilithium-3 signed
Vetos logged	25,386	Real-time veto tracking
Real Kraken trades	2,237	Phase 0 (Jul-Aug 2025)
Paper trades	119	Learning Baseline (Nov-Jan)

Metric	Value	Notes
Exit governance receipts	78	3-gate exit pipeline
B2B clients registered	5	With API key authentication
Sandbox sessions	3	Test sessions created
Sandbox evaluations	6	Evaluations run in sandbox
Vertical demos live	4	Trading, Credit, Insurance, Energy, Biotech
Governance checkpoints	8	CP-0 through CP-7 (fail-closed)
Compliance modules	5	NIST AI RMF / ISO 42001 / EU AI Act
Development months	~3	Dec 25, 2025 - Mar 13, 2026
Target event	Eureka Dubai GCC	March 18, 2026 — Semifinalist

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